

Claims status report

False, refuted, and incomplete statements

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This companion document records claims that must not be presented as settled theorems. It distinguishes a claim that has been checked and found false from one for which the available argument is incomplete, unsupported, or still an open research question.

The distinction is mathematically important:

- **False or corrected** means that a counterexample, direct calculation, or logical contradiction identifies a defect in the stated claim.
- **Incomplete or unsupported** means that the claim may still be true, but the available proof or certificate does not establish it.
- **Open** means that neither a complete proof nor a validated disproof is currently available in the project.

No item in this report is imported into Lean as an axiom. The report is a boundary document for the formalization: it says what the project will not silently treat as a theorem.

1. False or corrected claims

1.1 The explicit graph statistic is $2/3$, not $1/17$

For the displayed six-by-six bipartite graph, exhaustive enumeration gives:

- 18 perfect matchings in the full graph;
- 12 perfect matchings after deleting the selected edge (1,4);
- 8 witnesses for the selected nonincident vertex 12.

Consequently the witness fraction is

$$8/12=2/3,$$

not $1/17$. The minimum over the tested nonincident edge-vertex pairs is also $2/3$. The corrected statistic is the one retained in the final report.

1.2 Cubic bipartite does not imply three-edge-connectivity

Connected cubic bipartite multigraphs can have a two-edge cut. A small example has adjacency matrix

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

where the two single-copy edges form a two-edge cut. Bridgelessness, 2-edge-connectivity, 3-edge-connectivity, brace structure, and tight-cut irreducibility are different properties and cannot be substituted for one another.

1.3 An unguarded surplus formula is not valid

The surplus expression used in one historical branch omits a necessary right-degree condition. The resulting statement fails outside the regular setting. The corrected theory keeps the right-degree hypotheses explicit and does not use the unguarded expression as a theorem.

1.4 A general identity-minimizer assertion is false

The historical claim that a particular identity labeling is always the unique minimizer is not valid in the stated generality. The formalization uses support families and their potentials rather than that identity-minimizer assertion.

1.5 A Hamiltonicity inference from adding one edge is invalid

In the special situation $G=R-e$, adding the missing edge to a spanning path can create a Hamiltonian cycle. Therefore the statement that $G-M$ can be Hamiltonian while $R-M$ is not does not follow in that setting. The factor and complement conventions must be fixed before making a Hamiltonicity claim.

1.6 Higher parallel multiplicity is not an omitted cubic search class

In a cubic graph, an edge multiplicity greater than three is impossible, and three parallel copies isolate a two-vertex component. There is no connected cubic search family with an unbounded multiplicity that was omitted merely by stopping at multiplicity three.

2. Incomplete or unsupported claims

2.1 The main exact-list lower bound is not yet Lean-checked

The ordinary proof architecture for

$$N(L,R) \geq c_k(R)$$

is coherent after the common residual fiber and parallel-edge conventions are made explicit. The current Lean project has the data model and elementary support layer, but the complete residual path/cycle comparison, termination, and nested-count comparison remain formalization obligations.

2.2 The common-core converse remains open

The equality framework suggests that, in connected regular instances, equality should be characterized by a common core of size at least $k-1$. The available work reduces the difficult converse to TF_k ; it does not prove the converse.

2.3 Universal TF_k and cubic TF_3 remain open

The three-part spanning-2-factor property is a useful reduction target. No complete proof is available for all degrees, and no validated universal cubic counterexample has been supplied. Proposed gadgets without explicit graph data, exact matching certificates, and independently checkable witnesses do not constitute a disproof.

2.4 The small-shore proof outline is not a finished theorem

The latest small-shore reduction contains useful structural restrictions, but its final theorem is labeled a proof outline. Missing contraction hypotheses, port conventions, or terminal-case arguments cannot be filled by an earlier version that used a stronger theorem heading.

2.5 The Pfaffian branch has external dependencies

The Pfaffian cubic route is a coherent manuscript-level argument under the stated matching and brace inputs. It is not yet a Lean proof. Square-switch degree matching, pure-interface endpoint hypotheses, and the exact scopes of the external matching theorems must be formalized or proved locally before the branch can be called machine-verified.

2.6 General per-edge-list counting is a separate problem

The exact-list theorem assigns one list to each left vertex and shares it across that vertex's incident edges. Arbitrary per-edge lists are broader. The project does not prove that the same lower bound holds in that broader model.

2.7 The full matroid Latin-floor statement is not established

The purely parallel specialization has a graph interpretation and connects the complete bipartite case with Latin squares. That specialization does not prove the corresponding counting statement for arbitrary matroids.

2.8 Large historical computation claims lack complete certificates

Some reported large search totals do not include the graph corpus, source version, seed, command line, or executable output needed for reproduction. Some runs depend on missing modules or unavailable paths. Those totals remain historical research notes rather than finite certificates.

2.9 Port-mask reductions need exact hypotheses

The port-mask intersection criterion is valid as a reduction when the matched port, factor nonemptiness, and gluing convention are specified. It does not automatically imply that every brace has every desired mask, nor does it prove the existence of a global cubic obstruction.

3. Claims that are positively certified

The negative classifications above do not erase the project's positive results. The following are retained in the final report:

- finite edge-copy graph, exact-list, and support-family definitions;
- elementary list/support inverse identities in Lean;
- the explicit graph calculation 18, 12, 8, and 2/3;
- bounded independent enumeration results within their stated domains;
- the audited uncrossing proof architecture;
- the reduction framework connecting equality to TF_k ;
- the stated Pfaffian subclass route, with its external dependencies marked;
- the purely parallel graph/Latin-square translation;
- the exact list of remaining Lean proof obligations.

4. Formalization rule

The Lean development follows one rule: a claim enters the theorem layer only with an explicit definition and a proof term accepted by the kernel. A finite program output may be preserved as a certificate target, and a manuscript argument may guide the next formalization step, but neither is substituted for the proof term itself.

This rule keeps the final public report comprehensive without making it claim more than the mathematics and software currently establish.